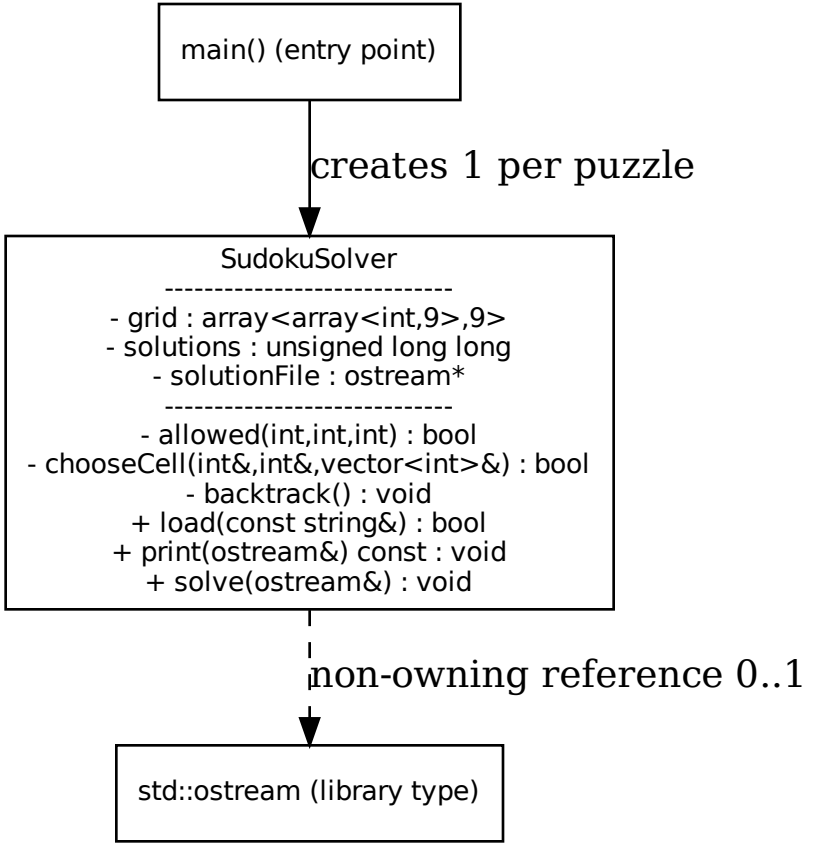
Assignment 6 — UML Class Model



Assignment 6 — UML Class Diagram

Implementation-specific class model

SudokuSolver

Private attributes: grid : std::array<std::array<int,9>,9>; solutions : unsigned long long; solutionFile : std::ostream*.

Private methods: allowed(row:int, col:int, value:int) : bool; chooseCell(bestRow:int&, bestCol:int&, bestValues:std::vector<int>&) : bool; backtrack() : void.

Public methods: load(filename:const std::string&) : bool; print(out:std::ostream&) const : void; solve(combinedOutput:std::ostream&) : void.

Relationships

SudokuSolver contains a 9×9 grid and a counter (composition/value members). SudokuSolver temporarily references std::ostream while solving (association; non-owning). The main() function creates a SudokuSolver instance for each puzzle and owns the std::ofstream that supplies the shared stream. There are no additional user-defined classes.

Method collaboration

main() → load() → allowed(); main() → print(); main() → solve() → backtrack() → chooseCell() → allowed(); backtrack() → print().